

## **1. Administrative & Study Identification**

**Public Study Title:** A Clinical Study of Zilovetamab Vedotin (MK-2140) Plus Rituximab Plus Cyclophosphamide, Doxorubicin, and Prednisone (R-CHP) Versus Polatuzumab Vedotin Plus R-CHP in People With Diffuse Large B-cell Lymphoma (DLBCL) (MK-2140-011/waveLINE-011) **Full/Official Study Title:** A Randomized, Open-label, Multicenter, Phase 2 Study Evaluating the Efficacy and Safety of Zilovetamab Vedotin (MK-2140) Plus R-CHP Versus Polatuzumab Vedotin Plus R-CHP in Treatment-naïve Participants With GCB Subtype of Diffuse Large B-cell Lymphoma (DLBCL) **Short Title/Acronym:** waveLINE-011 / MK-2140-011 **Protocol Number:** MK-2140-011 **NCT Number:** NCT06890884 **Posting ID:** 172471228181373134 **Sponsor/Company Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Zilovetamab vedotin (MK-2140) and Polatuzumab vedotin **Phase of Development:** Phase 2 **Trial Status:** Recruiting **Medical Monitor:** Clinical Director, Merck Sharp & Dohme LLC

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## **2. Study Synopsis & Rationale**

**2.1 Study Objective Primary Objective:** To evaluate if first-line treatment with zilovetamab vedotin (MK-2140) plus R-CHP increases the Complete Response Rate (CRR) at the end of treatment (EOT) compared to polatuzumab vedotin plus R-CHP. **Secondary Objectives:** To evaluate progression-free survival (PFS), overall survival (OS), and event-free survival (EFS).

**2.2 Background & Rationale** Diffuse large B-cell lymphoma (DLBCL) is a fast-growing hematologic malignancy. The germinal center B-cell-like (GCB) subtype specifically affects young, maturing B-cells. Novel, first-line treatment options for GCB DLBCL are required. Receptor tyrosine kinase-like orphan receptor 1 (ROR1) is an oncofetal transmembrane protein overexpressed in DLBCL. Zilovetamab vedotin is an investigational ROR1-directed antibody-drug conjugate (ADC). This study aims to determine whether adding zilovetamab vedotin to standard R-CHP therapy provides superior tumor response compared to the active comparator polatuzumab vedotin plus R-CHP.

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## **3. Study Design & Methodology**

**3.1 Design Framework Study Type:** Interventional **Control Type:** Active Comparator (Polatuzumab vedotin) **Blinding:** Open-label **Randomization:** Randomized

**3.2 Planned Enrollment & Duration Targeted Enrollment:** 594 participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 11-Apr-2025 **Estimated Primary Completion:** 13-Dec-2027

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## **4. Subject Selection (Eligibility Criteria)**

**4.1 Inclusion Criteria** 1. Has histologically confirmed diagnosis of germinal center B-cell (GCB) subtype of diffuse large B-cell lymphoma (DLBCL), by prior biopsy, according to the World Health Organization (WHO) classification of neoplasms of the hematopoietic and lymphoid tissues. 2. Has positron emission tomography (PET) positive disease at screening, defined as 4 to 5 on the Lugano 5-point scale. 3. Treatment-naïve: Has received no prior treatment for their DLBCL. 4. Human immunodeficiency virus (HIV) infected participants must have well controlled HIV on antiretroviral therapy (ART). 5. Participants who are hepatitis B surface antigen (HBsAg) positive are eligible if they have received hepatitis B virus (HBV) antiviral therapy and have undetectable HBV viral load prior to randomization. 6. Participants with history of hepatitis C virus (HCV) infection are eligible if HCV viral load is undetectable at screening.

**4.2 Exclusion Criteria** 1. Has a history of transformation of indolent disease to DLBCL. 2. Has received a diagnosis of primary mediastinal B-cell lymphoma (PMBCL) or Grey zone lymphoma. 3. Has Ann Arbor Stage I DLBCL. 4. Has clinically significant (i.e., active) cardiovascular disease: cerebral vascular accident/stroke (<6 months prior to enrollment), myocardial infarction (<6 months prior to enrollment), unstable angina, congestive heart failure (New York Heart Association Classification Class ≥II), or serious cardiac arrhythmia requiring medication. 5. Has clinically significant pericardial or pleural effusion. 6. Has ongoing Grade >1 peripheral neuropathy. 7. Has a demyelinating form of Charcot-Marie-Tooth disease. 8. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 9. Has ongoing corticosteroid therapy. 10. Known additional malignancy that is progressing or has required active treatment within the past 2 years. 11. Known active central nervous system (CNS) lymphoma. 12. Has active autoimmune disease that has required systemic treatment in the past 2 years. 13. Has active infection requiring systemic therapy. 14. Has active HBV (defined as HBsAg positive and detectable HBV deoxyribonucleic acid (DNA)) and HCV (defined as anti-HCV antibody positive and detectable HCV ribonucleic acid (RNA)) infection. 15. Has history of stem cell/solid organ transplant.

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## **5. Intervention & Treatment Plan**

**5.1 Investigational Regimen Dosage/Frequency:** Zilovertamab vedotin (MK-2140) or Polatuzumab vedotin administered in

combination with Rituximab, Cyclophosphamide (Trade names: Cytosan, Endoxana; ATC code: L01AA01), Doxorubicin, and Prednisone (R-CHP) in specific treatment cycles. **Route of Administration:** Intravenous (IV) infusions for biologics/chemotherapy; Oral for Prednisone. **Duration of Treatment:** Until completion of protocol-specified cycles, disease progression, or unacceptable toxicity.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard supportive oncology care; Antiretroviral therapy for HIV; Antiviral therapy for HBV (if viral load is undetectable). **Prohibited:** Prior systemic treatments for DLBCL; Concurrent investigational agents.

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days) | Key Activities/Procedures                                                      |
|--------------|---------------|--------------------------------------------------------------------------------|
| Screening    | Day -28 to 0  | Informed Consent, Medical History, Tumor Biopsy/PET-CT, Viral Labs             |
| Baseline     | Day 0         | Randomization, First Dose Administration                                       |
| Interim      | Every Cycle   | Safety Labs, Efficacy Assessment (PET-CT), Adverse Event Monitoring            |
| End of Study | Follow-up     | Final Vital Signs, Exit Interview, Survival Follow-up (until approx. Dec 2032) |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Complete Response Rate (CRR) at End of Treatment (EOT) per Lugano Response Criteria. **Measurement Method:** CRR at EOT is defined as the percentage of participants who experience complete response (CR) per Lugano response criteria as assessed by blinded independent central review (BICR) at end of treatment. CR is complete metabolic (no/minimal FDG uptake) and radiologic response (target lesions regress to  $\leq 1.5$  cm in longest transverse diameter of a lesion) and no new lesions. Participants with missing data or who discontinue treatment or study prior to reaching EOT will be considered non-responders and included in the total number of participants.

**7.2 Secondary & Exploratory Endpoints** \* Progression-free survival (PFS) per Lugano Response Criteria \* Overall survival (OS) \* Event-free survival (EFS) per Lugano Response Criteria

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous collection via patient reporting, physical exams, and lab assessments at every cycle. **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee**

(DMC): Independent Data Monitoring Committee established to routinely evaluate unblinded safety and efficacy data.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy analyses; Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** Target  $N=594$  powered appropriately to detect a statistically significant difference in CR rates between the zilovertamab vedotin and polatuzumab vedotin study arms. **Handling of Missing Data:** Participants with missing data or who discontinue treatment or study prior to reaching EOT will be considered non-responders.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Combination of onsite and remote monitoring to ensure subject safety, protocol adherence, and data integrity. **Audit Trail:** Validated Electronic Data Capture (EDC) system utilizing automated audit trails and data clarification forms.

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## 11. Study Identifiers & Governance

**Primary ID:** MK-2140-011 **Registry Number:** NCT06890884 **Posting ID:** 172471228181373134 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** waveLINE-011 **Study Preferred UMLS Name:** Diffuse Large B-cell Lymphoma **Target Terminology Identifiers:** MeSH Code: D001402 (B-Lymphocytes / B-Cell) | SNOMED ID: 112130006 **Disease Areas:** Lymphoma, Large B-Cell, Diffuse

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## 12. Clinical Framework (Oncology Specific)

**Primary Condition:** Diffuse Large B-cell Lymphoma (GCB Subtype) **Diagnosis Threshold:** PET positive disease defined as 4 to 5 on the Lugano 5-point scale **Medical Methodology:** Antibody-Drug Conjugate (ADC) combined with Chemotherapy **Optimization Target:** Complete Response (CR)

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## 13. Study Procedures & Methodology

**Management Pathway:** Standard first-line oncology clinical pathway for DLBCL **Education Protocol (HCP):** Protocol-specific training, site initiation visits, and investigator meetings **Education Protocol (Patient):** Oncology treatment education and thorough

informed consent counseling **Quality Audit Mechanism:** Centralized data monitoring and source data verification

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Up to 7.5 years (Est. Study Completion Dec 2032)

**Onsite Visit Cadence:** Day 1 of each treatment cycle and regularly scheduled follow-ups **Remote Monitoring Frequency:** Continuous EDC data review **Checklist Review Interval:** Prior to administration of each dosing cycle

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## 15. Sign-off & Approval

| Role                     | Name   | Signature | Date          |  | :--- | :--- | :--- | :--- |
|--------------------------|--------|-----------|---------------|--|------|------|------|------|
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |  |      |      |      |      |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |  |      |      |      | Lead |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |  |      |      |      |      |